

Long Short-Term Memory (LSTM) Networks

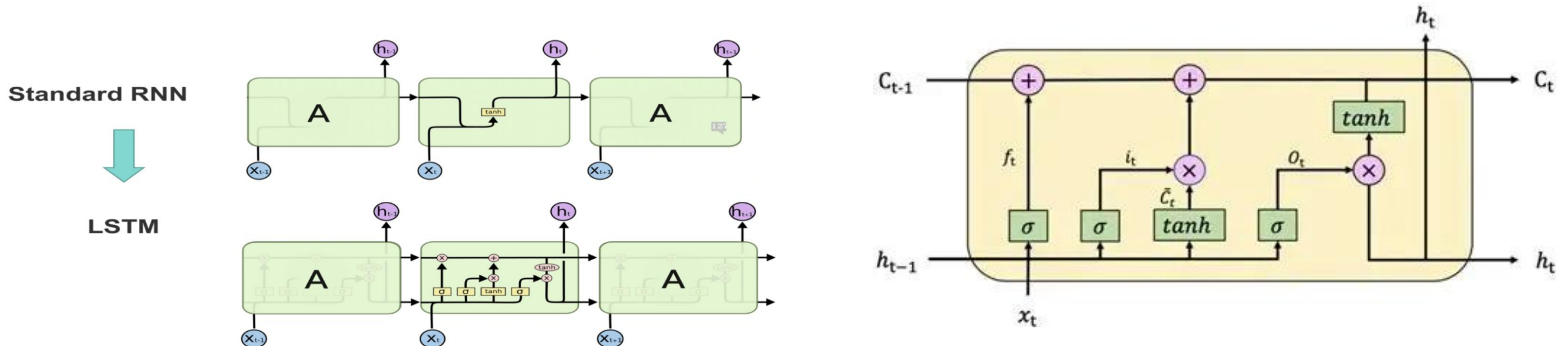
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Lecture 14

ANN & DL

Definition

- LSTM (Long Short-Term Memory) is a special type of Recurrent Neural Network (RNN) designed to **learn long-term dependencies** by controlling information flow using **gates and memory cells**.
- Standard RNN $\rightarrow$ short memory
LSTM $\rightarrow$ long + controlled memory

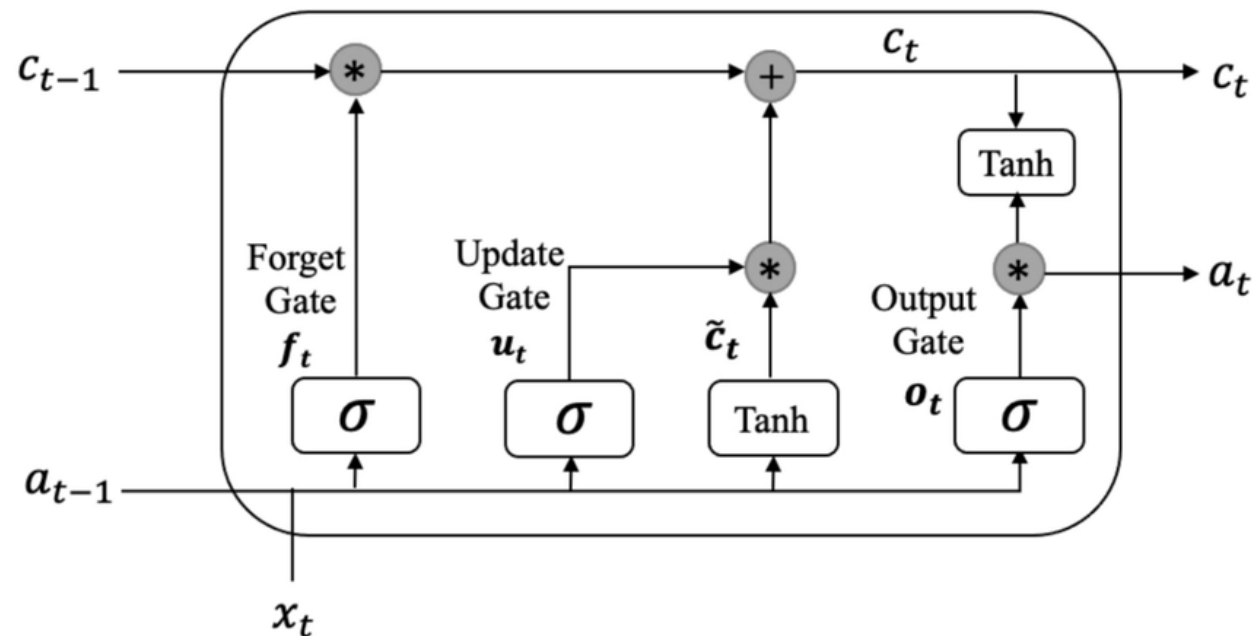


Why Not CNN or Simple RNN?

- **CNN** Best for: **images**, spatial data
- Limitation: Cannot model **sequential/time dependencies**
- Example: CNN can detect objects in a frame.
- But cannot understand **sentence meaning over time**
- **RNN** Designed for **sequences**
- **But** suffers from **Vanishing gradient problem**, **Cannot remember** long sequences.
- Gradients shrink → **learning stops**
- Model **forgets** earlier information

LSTM

- LSTM solves Long-term memory problem by introducing memory cells (gating mechanism)
- Human brain Remembers important things, Forgets irrelevant details.
- LSTM introduces **Cell state (memory line)** in 3 gates:
 - (1) Forget Gate
 - (2) Input Gate
 - (3) Output Gate



LSTM ARCHITECTURE

- Components of LSTM

(1) Cell State C_t Long-term memory , Flows through network

(2) Forget Gate Decides What to **forget** from previous memory

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

(3) Input Gate Decides What new information to **store**

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

(4) Candidate Memory $\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$

(5) Update Cell State $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$

(6) Output Gate $o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$

(7) Final Output $h_t = o_t \cdot \tanh(C_t)$

Interpretation

Gate

Forget Gate

Input Gate

Output Gate

Role

Remove useless info

Add important info

Decide what to show

Applications of LSTM

- Speech recognition
- Machine translation
- Time-series forecasting
- Medical diagnosis (ECG signals)
- NLP (chatbots, sentiment analysis)

Q.1. Given At time step t : Previous cell state: $C_{t-1} = 0.5$

Forget gate: $f_t = 0.8$, Input gate: $i_t = 0.6$, Candidate memory: $\tilde{C}_t = 0.7$

Find New cell state C_t .

Solution:

Using formula $C_t = f_t C_{t-1} + i_t \tilde{C}_t$

Substituting the given values into the formula

$$C_t = (0.8 \times 0.5) + (0.6 \times 0.7)$$

$$C_t = 0.4 + 0.42 = 0.82$$

Q.2. From the previous question use $C_t = 0.82$ and if the Output gate return $o_t = 0.9$ Find the Hidden state h_t

Solution:

$$h_t = o_t \cdot \tanh(C_t)$$

$$\tanh(0.82) \approx 0.675$$

$$h_t = 0.9 \times 0.675 = 0.6075$$

So the hidden state remember $h_t \approx 0.61$ (61%) of the information

Q3. A model processes the sentence:

“The patient was stable for days but suddenly developed severe infection.”

Why would RNN fail here? How does LSTM help?

Solution:

RNN Issue is that it cannot remember **“stable for days”**

focuses only on recent words (**“severe infection”**)

LSTM Advantage:

Keeps long-term memory connects earlier context → later outcome

That cause better prediction of **disease progression**

Q.4. You are predicting patient heart rate trends. Why is LSTM better than CNN?

Solution:

Model	Limitation
CNN	No temporal memory
RNN	Short memory
LSTM	Captures long-term trends

Q.5. You have given $f_t = 0, i_t = 1$ if you start to solving this problem what happens during the solution?

Solution:

During solution it forget everything from past.

Fully accept new information, Model behaves like fresh memory reset.

Case Study: A hospital uses an LSTM model to monitor a patient's infection severity over time. At time t , the system has Previous severity memory $C_{t-1} = 0.6$, Forget gate $f_t = 0.8$ retain most past information. Input gate $i_t = 0.4$ moderate new information. New candidate severity $\tilde{C}_t = 0.9$. **What is the updated severity level C_t ? Also Interpret the result.**

Solution:

Step 1: Apply LSTM update rule $C_t = f_t C_{t-1} + i_t \tilde{C}_t$

Step 2: Substitute values $C_t = (0.8 \times 0.6) + (0.4 \times 0.9)$

Step 3: Calculate $C_t = 0.48 + 0.36 = 0.84$

Interpretation:

Past condition strongly retained. New infection signal increases severity.

Patient condition is **worsening**

Assignment

Case Study-1: An LSTM predicts stock trends. Yesterday's trend is strong. Previous trend memory $C_{t-1} = 0.7$. Forget gate $f_t = 0.9$, Input gate $i_t = 0.2$, New market signal $\tilde{C}_t = 0.3$. Find the new trend value and explain in your own words.

Case Study-2: An LSTM processes the sentence. *"The service was initially bad but later became excellent."* At the word "excellent", the network has previous sentiment memory $C_{t-1} = -0.5$ negative, forget gate $f_t = 0.3$ and Input gate $i_t = 0.9$ whereas candidate sentiment $\tilde{C}_t = 1.0$. Compute final sentiment C_t and interpret the results in your own words.

Case Study-3: An LSTM predicts temperature changes over time (t). Previous temperature memory $C_{t-1} = 20$, Forget gate $f_t = 0.85$, Input gate $i_t = 0.3$, whereas the new weather input $\tilde{C}_t = 25$. Find updated temperature state C_t .

Case Study-4: An LSTM model predicts disease risk of patients suffering of particular disease. Cell state $C_t = 1.2$, Output gate $o_t = 0.6$, compute the value at hidden state h_t and interpret the results in simple words so that doctor can understand the disease severity.

Case Study-5: An LSTM processes a sudden change event on a mobile device. If $C_{t-1} = 0.9$, $f_t = 0.2$, $i_t = 0.8$, $\tilde{C}_t = 0.4$. Compute C_t and explain the behavior of the device.

Thank you